Moles & reacting masses

Count accurately

1 M_r & composition

$Ca(OH)_2$: 40 + 2 × (16 + 1) = **74.** Brackets multiply every atom inside.

Oxygen percentage = 32/74 × 100 = **43.2%.** M_r has no unit; molar mass is 74 g/mol.

2 Moles & particles

n = mass / molar mass. 11 g CO_2 / 44 g/mol = **0.25 mol.**

Particles = n × 6.02 × 10^{23}. This is 1.505 × 10^{23} CO_2 molecules; each has three atoms.

State whether counting atoms, molecules, ions or electrons.

Follow the recipe

3 Reacting masses

mass | moles | ratio

Convert → compare → convert back

$2Mg + O_2 \rightarrow 2MgO.$ 6.0 g Mg / 24 = 0.25 mol Mg.

Mg: MgO = 2: 2 → 0.25 mol MgO. Mass = 0.25 × 40 = **10 g MgO,** oxygen in excess.

Coefficients are mole ratios, not mass ratios.

4 Find equation coefficients

Divide each mass by its molar mass; divide all moles by smallest; multiply to whole numbers.

2.4 g Mg, 1.6 g O_2, 4.0 g MgO → 0.10: 0.05: 0.10 mol → **2: 1: 2.**

Check atoms and total mass on both sides.

Check what runs out

5 Limiting vs excess

0.30 mol Mg + 0.10 mol O_2: oxygen needs only 0.20 mol Mg.

O_2 is limiting. MgO formed = **0.20 mol;** Mg left = **0.10 mol.**

Use limiting amount for theoretical yield; extra excess reactant cannot increase it.

6 Mass & uncertainty

An open vessel gains mass if oxygen enters; loses mass if gas escapes. Count all substances to conserve mass.

Readings 24.2, 24.4, 24.6 → mean 24.4; range 0.4; half-range uncertainty **±0.2.**

Percentage uncertainty = 0.2/24.4 × 100 = 0.82%.